

Designing the system

Fig. 2 shows the circuit diagram of the integrated motor-driver robot chassis system. The system is built around the DRV8833 dual H-bridge motor-driver IC, which provides independent control of two N20 DC geared motors while integrating power distribution and sensor connectivity on a single PCB.

Power is supplied via the screw terminal (CON1), while a 24-pin FPC connector serves as the primary interface to an external controller, such as an ESP32 or an Arduino. The FPC cable carries motor-control signals (IN1-IN4), I²C communication lines (SDA and SCL), GPIO signals, sensor inputs, and power rails, greatly simplifying wiring and assembly.

During operation, the controller generates logic and PWM signals that are applied to the DRV8833 inputs. The motor driver converts these signals into high-current outputs to drive the two motors in forward, reverse, brake, or stop modes, enabling precise control of the robot's speed and direction. Additional connectors are provided for sensor integration and future expansion, making the platform suitable for a wide range of robotics applications.

The system employs the DRV8833PW dual H-bridge motor driver in a compact HTSSOP package. The device operates from a motor-supply voltage of 2.7V to 10.8V and can deliver up to 1.2A of continuous current per channel, with peak currents of up to 2A. Motor speed is controlled using PWM signals, while motor direction is determined by the logic states applied to the input pins. The IC also includes a low-power sleep mode (nSLEEP) and a fault output (nFAULT) that provides protection against over-current, over-temperature, and under-voltage conditions.

As shown in Fig. 2, the 10µF, 2.2µF, and 10nF capacitors provide

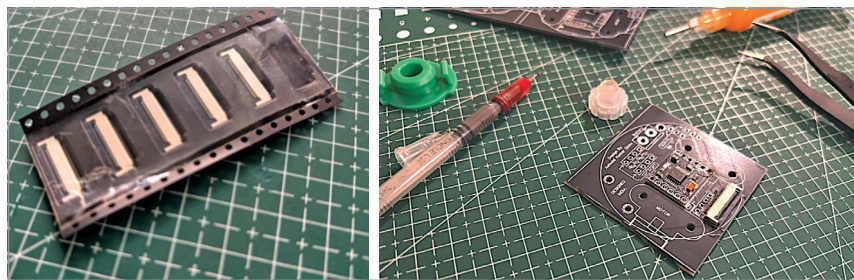


Fig. 4: Fabricated PCB and completed SMT assembly

bulk decoupling, supply filtering, and charge-pump stabilisation, ensuring reliable operation of the motor driver. The main motor supply is connected through CON1 and distributed via the VIN rail. A solder jumper allows VIN to be linked to the 3.3V rail, enabling both the motors and the controller to be powered from a single battery when required.

The PCB also includes dedicated servo connectors, with signal lines routed from available GPIO pins and power supplied directly from the VIN rail. Two I²C expansion connectors are provided for easy integration of modules such as IMUs, distance sensors, environmental sensors, OLED displays, and other I²C-compatible peripherals.

By integrating motor control, power distribution, sensor interfaces, and mechanical mounting features onto a single PCB, the system provides a compact, modular, and easy-to-assemble robotic platform with minimal wiring requirements.

PCB design

After finalising the schematic (Fig. 2), the PCB layout was developed. The DRV8833 motor driver is positioned near the centre of the board to optimise signal routing and ensure efficient power distribution. The N20 motors are mounted

```

3 // With LED Blink on Pin 33 on every movement change
4 // =====
5
6 #define IN1 1 // Motor A
7 #define IN2 3
8 #define IN3 4 // Motor B
9 #define IN4 5
10
11 #define LED_PIN 33
12
13 const int freq = 5000;
14 const int resolution = 8;
15
16 // PWM Channels
17 const int chIN1 = 0;
18 const int chIN2 = 1;
19 const int chIN3 = 2;
20 const int chIN4 = 3;
21
22 void setup() {
23   Serial.begin(115200);
24   Serial.println("DRV8833 Test - ESP32 Core 2.0.17 Version");
25
26   // Set pins as output
27   pinMode(IN1, OUTPUT);
28   pinMode(IN2, OUTPUT);
29   pinMode(IN3, OUTPUT);
30   pinMode(IN4, OUTPUT);
31   pinMode(LED_PIN, OUTPUT);
32   digitalWrite(LED_PIN, LOW);
33
34   // LEDC Setup for Core 2.0.17
35   ledcSetup(chIN1, freq, resolution);
36   ledcSetup(chIN2, freq, resolution);

```

Fig. 5: Snippet of the source code

directly onto the chassis PCB, with dedicated mounting slots and solder pads provided for secure mechanical and electrical connections.

A solder jumper, as shown in the circuit diagram, enables flexible power configuration by allowing users to either share a common battery supply between the motor driver and microcontroller or keep the two power domains isolated.

The servo connectors are positioned along the upper-right section of the chassis for convenient access, while the I²C expansion connectors are located near the top centre of the board. Multiple mounting holes are provided at the front and rear of the chassis to support stackable expansion boards and additional